Evaluation Model Comparison and Final Decision

Worst-Case vs. Gaze-Weighted — Findings, Defects, and Recommendation

IYPT Pinhole Sunglasses — Array Geometry Sub-Study

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1 The Two Evaluation Models

Model A — Worst-case gates (this project's original primary standard). A design must remain ghost-free and non-blind at every tested gaze eccentricity φ from 0° to 25°. Pass/fail, no weighting. Ranks survivors by effective density.

Model B — Gaze-weighted (added later). Integrates the measured eccentricity distribution over φ, sampling 36 azimuths per eccentricity bin, and reports the *fraction of actual viewing time* a design spends blind or ghosting.

The two were built to differ in one intended dimension: whether rare peripheral angles count as heavily as the frequently-used centre. They turned out to differ in a second, unintended dimension as well — and that second difference is the important finding of this document.

2 Critical Finding: Model A Is Not Actually Worst-Case

Model A samples the pupil shift along a **single azimuthal direction** ($+x$) as φ increases. Model B samples 36 azimuths at each eccentricity. Because holes are discrete objects with gaps between them, a pupil can be fully lit when shifted in one direction and completely dark when shifted the same distance in another.

Verified directly on the M=6 gaze-weighted winner (`hexagonal`, N=100, pitch=4000 μm, a_{hole}=500 μm):

Sampling method	Minimum illuminated area
Single-axis ($+x$ only), as Model A uses	181,457 μm^2 (never blind)
All 36 azimuths, as Model B uses	**0 μm^2 (blind at 3 of 936 samples)**

Critical Finding

Across the full 8818-row candidate set, **1750 rows (19.8%)** are reported as "never blind" by Model A's `min_illuminated_area_across_phi` while Model B's azimuthal sampling finds genuine blindness — in the worst case, up to **20.3% of viewing time**. This affects *every* geometry family tested (jittered_lattice 641, rectangular 493, poisson_disk 154, square 124, hexagonal 54, and all others). Model A is therefore **not strictly more conservative than Model B**: it is stricter in eccentricity (treating 25° as heavily as 0°) but blind in azimuth.

3 Findings Per Model

3.1 Model A — Worst-Case Gates

Strengths. Simple, conservative in eccentricity, produces hard pass/fail guarantees. Correctly rejected every genuinely broken design found during this project's audits (blackout candidates, boundary near-misses, seed-lottery winners).

Defects found.

- **Single-azimuth sampling** (Section 2) — misses blind spots in 19.8% of candidates. This is the most serious defect in either model.
- **Density objective reduces to 1/hull area** when only one hole is active (true for M=2–8), so it effectively minimises array size rather than maximising light. It also scales with $N_{\text{requested}}$, systematically favouring the smallest tested N — every Model A winner sits at the minimum N swept.
- **Unweighted eccentricity** — treats 25° (0.58% of viewing time) as heavily as 0° (9.29%), disqualifying designs on failures the wearer would essentially never encounter.

3.2 Model B — Gaze-Weighted

Strengths. Azimuthally complete. Weights by measured viewing-time distribution (verified: weights sum to 100.0000, cumulative 51.61% by 6° and 95.27% by 17°, all three assertions passing). Outputs a directly interpretable quantity ("% of viewing time ghosting") rather than an abstract margin.

Defects and limitations.

- **Large tied plateaus** — many designs reach exactly 0.00%/0.00%, so the reported "winner" is one representative of a tie chosen by insertion order, not a uniquely optimal design. The comparison report states this explicitly rather than implying a sharper distinction than the data supports.
- **No density or light-throughput term** — it optimises purely for safety, so among equally-safe designs it has no basis to prefer the one delivering more light.
- **Depends on one empirical distribution** — a different gaze dataset would shift the weights, though the sharp central peak is robust across the literature.

4 Result Comparison

M	Model A winner	Model B winner	Model B ghost/blind
0	hexagonal, N=20, p=2000,30°	square, N=20, p=4000	0.00% / 0.00%
1	ring_offset_conc., N=20, rs=4000	hexagonal, N=20, p=4000	0.00% / 0.00%
2–3	ring_offset_conc., N=20, rs=4000	hexagonal, N=20, p=4000	0.00% / 0.00%
4–5	ring_offset_conc., N=20, rs=4000	hexagonal, N=20, p=4000	0.00% / 0.00%
6	ring_offset_conc., N=20, rs=4000	hexagonal, N=100, p=4000	0.00% / 0.12%
8	ring_offset_conc., N=20, rs=4000	hexagonal, N=50, p=4000	0.00% / 0.33%
global	hexagonal, N=20, p=2000,30°	square, N=20, p=4000	0.00% / 0.00%

Every Model B winner also carries $+\infty$ worst-case ghost and vignetting margin under Model A — verified directly. The disagreement is not about safety; both sets are safe under Model A's own gates. The models simply optimise different objectives among safe candidates (density vs. time-weighted safety).

Structural difference in the winners. Model A's winners cluster at dense spacings across three different families. Model B's winners are uniformly the *sparsest* tested pitch (4000 μm), overwhelmingly one family (hexagonal). Sparse layouts keep holes far enough apart that no azimuth at any eccentricity brings two holes simultaneously active, while the 4 mm pupil remains large enough that blackouts stay rare.

5 Decision

Decision

Use Model B (gaze-weighted) as the primary evaluation model, with Model A's ghost and vignetting margin gates retained as a mandatory pre-filter.
Rationale: Model A's single-azimuth sampling is a genuine correctness defect affecting 19.8% of candidates — it is not merely more conservative, it is incomplete in a dimension Model B covers. Model B is azimuthally complete, weighted by measured viewing behaviour, and outputs an interpretable quantity. But Model B alone has no hard safety floor, so Model A's margin gates must still be applied first to exclude designs that fail catastrophically at any angle.

Practical procedure: filter candidates by Model A's ghost and vignetting margin gates (both $>$ threshold), then rank the survivors by Model B's time-weighted ghost % with blind % as tie-break. All nine current Model B winners already satisfy both conditions.

5.1 Caveat to State in Any Report

Model A's `min_illuminated_area_across_phi` column should not be cited as evidence a design is never blind — Section 2 shows it is unreliable for that purpose. Use Model B's `time_weighted_blind_pct` instead. Model A's *margin* columns are unaffected by this defect and remain valid.